Session 11

Guiding principles and best practices for data capacity building

Ernest Guevarra

ernest@guevarra.io

University of Oxford

Building data capacity

- Not everyone needs to be a data scientist.
- Data capacities in policymaking refer to how well infrastructure, systems, and people can use data effectively for decisions.
- Strengthening these abilities across different teams is essential.
- Complementary skill sets help reinforce the entire data value chain and improve overall decision-making.

1. Identify gaps in data skills and statistical capacities

Types of skills

Technical know-how

Mechanical, ICT, mathematical, scientific and/or other technology-related skills such as proficiency with a standard statistical package (e.g., R; Stata; SPSS) to carry out data processing, analysis and dissemination.

Work know-how

An employee's knowledge and understanding of work practices, processes, statistical concepts, definitions and classifications, business models, and organisational culture in use in their organisation, as well as about the power relationships within and between organisations.

Problem-solving and creative thinking

The traits that support defining, approaching and analysing a problem in a technically correct manner and from a fresh perspective to devise new ways for carrying out tasks, meeting challenges and solving problems.

Data Capability Framework

	PLAN	COLLECT	DESCRIBE	STORE	ANALYSE	USE	SAVE/DESTROY
Employ data coding and classification principles	□	□	■	■	□	■	■
Integrate data	□	□	■	□	■	■	□
Use data processing methodologies	□	□	□	□	■	□	□
Contribute to data outputs, products or service production	■	□	□	□	□	□	□
Perform exploratory data analysis	□	□	□	□	■	□	□
Conduct business intelligence data analysis	□	□	□	□	■	■	□
Conduct statistical data analysis	□	□	□	□	■	■	□
Conduct specialist data analysis	□	□	□	□	■	■	□
Identify and understand data availability	■	■	□	□	□	□	□
Employ data collection methodology	□	■	□	□	□	□	□
Contribute to data access design	■	■	□	□	□	□	□
Contribute to the sourcing and use of administrative data	■	■	■	□	□	■	□
Understand and contribute to data collection process design	■	■	□	□	□	□	□
Describe and summarise data	□	□	■	□	□	□	□
Understand and apply data editing methods	□	□	■	■	■	□	□
Use data quality assurance measures	□	■	■	□	■	■	□
Identify and evaluate data intelligence	■	■	■	□	□	■	□
Improve data processes/systems/products	■	□	□	■	□	■	■
Identify research questions	■	■	□	□	□	■	■
Apply data governance guidance	■	■	■	■	□	□	■
Value organisational data as assets	□	□	■	■	□	□	■
Employ statistical concepts and methodologies	□	■	■	■	■	■	■
Enable others to use and re-use data	□	□	■	■	□	■	■
Employ data and information management concepts	□	■	■	■	■	■	■
Visualise data	□	□	□	□	□	■	□

Data Capability Framework - example

- **Capability:** Employ data coding and classification principles
- **Catgory:** Describe, Store, Use, and Save/Destroy

New

- Is aware of relevant data classifications and coding protocols, and their proper application to data in general.
- Knows who to consult for expert knowledge.

Proficient

- Has a comprehensive knowledge of data classifications and coding protocols.
- Knows where to obtain expert advice about coding and classifications as needed.

Expert

- Is consulted regularly by others about data classifications and coding protocols.
- Can employ conceptual frameworks in support of data classification and coding.

Data Capability Framework Guide

Link

Data Maturity Assessment Framework

Topic: Engaging with others

Topic: Having the right data skills and knowledge

Topic: Having the right systems

Topic: Knowing the data you have

Topic: Making decisions with data

Topic: Managing and using data ethically

Topic: Managing your data

Topic: Protecting your data

Topic: Setting your data direction

Topic: Taking responsibility for data

Data Maturity Assessment Framework

- **Themes:** Uses, Data, Leadership, Culture, Tools, Skills
- **Maturity Levels:** Beginning, Emerging, Learning, Developing, Mastering

Data Maturity Assessment Framework - example

Topic: Engaging with others

ID	Row summary	Theme	Level 1: Beginning	Level 2: Emerging	Level 3: Learning	Level 4: Developing	Level 5: Mastering
1	Making data available to those who need it.	**Culture**	Makes data available by default only to a single specialist person or team. Discourages sharing data internally.	Beginning to share data internally either verbally or via reports. Does not encourage data sharing across teams or provide ways to share data directly.	Enables specialist users to access and share some data internally, but systems or processes limit access to and sharing of data by default rather than by design. Some internal users have appropriate access to data they need but may require specialist support to access or share data.	Beginning to provide ways to access and share data directly, but non-expert staff may require some intervention from specialists to do so. All internal users and some external users have appropriate access to data they need when they need it.	Data can be accessed and directly shared appropriately by all users who need it. All internal and external users can access data they need when they need it, without specialist support.

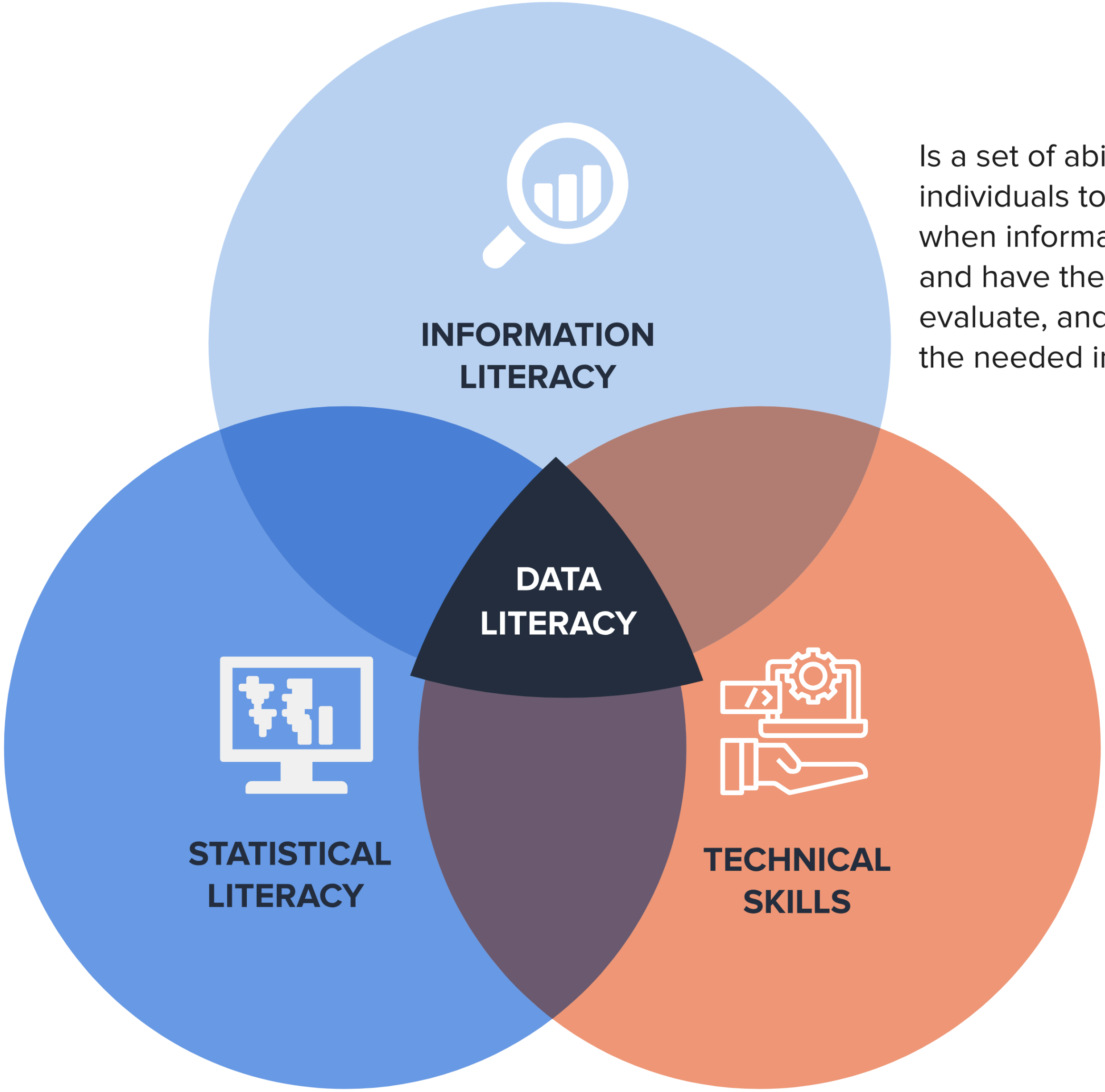
Data Maturity Assessment Framework Guide

Link

2. Build data capacities in your team

Data Literacy

Is a set of abilities requiring individuals to “recognize when information is needed and have the ability to locate, evaluate, and use effectively the needed information.



Refers to use of statistics as evidence, encompassing what to count or measure, how to assemble those measurements into summary statistics, what comparisons to form from these statistics and how to communicate these statistics.

Refer to skills that are essential for handling, analyzing, interpreting, and deriving insights from data, which has become increasingly important in the digital age.

Source: UN Data Revolution Group

Data Skills Framework

3. Operationalise data capacity building

Short-term

- Training workshops
- Online training resources
- Knowledge sharing on global innovations and use cases

Medium-term

- Internal data champions
- Periodic data skills training
- Pilot data projects
- Build complementary skills with other stakeholders
- Build on local knowledge and capacities
- Build network of data practitioners across your organization

Long-term

- Embed data skills in job roles
- Institutionalize data training programs
- Partner with think tanks and external organizations
- Partner with academia and local universities